

MultiView.LOCUS Analysis for Three Simulation Datasets

Contents

Install the package	1
(1) Helper functions - hidden	2
(2) Data Loading	2
Simulation Study 1 (simu_1) — tune phi/psi then final result	4
Simulation Study 2 (simu_2) — tune phi/psi then final result	6
Simulation Study 3 (simu_3) — show results only (no tuning)	8

The Rmd File is self-contained, no need to download data files separately.

Install the package

```
if (!requireNamespace("devtools", quietly = TRUE))
  install.packages("devtools")
if (!requireNamespace("MultiView.LOCUS", quietly = TRUE)) {
  devtools::install_github("Emory-CBIS/MultiView.LOCUS")
}## Additional packages used in this report
for (pkg in c("lattice", "gridExtra", "grid")) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg, repos = "https://cloud.r-project.org")
  }
}

library(MultiView.LOCUS)

## Loading required package: quadprog
## Loading required package: far
## Loading required package: nlme
## far library : Modelization for Functional AutoRegressive processes
## version 0.6-6 (2022-08-12)
## Loading required package: dplyr
## Warning: package 'dplyr' was built under R version 4.2.3
##
## Attaching package: 'dplyr'
## The following object is masked from 'package:nlme':
##
##     collapse
## The following objects are masked from 'package:stats':
```

```
##
## filter, lag
## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
## Loading required package: MASS
##
## Attaching package: 'MASS'
## The following object is masked from 'package:dplyr':
##
## select
## Loading required package: ica
## Loading required package: ggplot2
## Warning: package 'ggplot2' was built under R version 4.2.3
## Loading required package: lattice
## Loading required package: gridExtra
##
## Attaching package: 'gridExtra'
## The following object is masked from 'package:dplyr':
##
## combine
## Loading required package: reshape2
## Loading required package: tibble
library(lattice)
library(gridExtra)
library(grid)
```

(1) Helper functions - hidden

Functions for plotting and parameter selections

(2) Data Loading

This report runs the same tuning + final model workflow for:

- `simu_1`: `Data_x_simu1.rds` + `Data_y_simu1.rds`
- `simu_2`: `Data_x_simu2.rds` + `Data_y_simu2.rds`
- `simu_3`: `Data_x.rds` + `Data_y.rds`

```
load_rds_from_sources <- function(filename,
                                   package = "MultiView.LOCUS",
                                   urls = character(0),
                                   local_fallback = character(0)) {
  # 1) Installed package extdata
  pkg_file <- system.file("extdata", filename, package = package)
  if (nzchar(pkg_file) && file.exists(pkg_file)) {
    return(readRDS(pkg_file))
  }
}
```

```

# 2) URL(s)
for (u in urls) {
  obj <- tryCatch({
    con <- url(u, open = "rb")
    on.exit(close(con), add = TRUE)
    readRDS(con)
  }, error = function(e) NULL)

  if (!is.null(obj)) {
    return(obj)
  }
}

# 3) Local fallback path(s)
for (p in local_fallback) {
  if (file.exists(p)) {
    return(readRDS(p))
  }
}

stop("Could not load ", filename,
     " from package extdata, URL, or local fallback paths.")
}

repo_raw <- "https://raw.githubusercontent.com/Emory-CBIS/MultiView.LOCUS/main"

sim_data <- list(
  simu_1 = list(
    Data_x = load_rds_from_sources(
      "Data_x_simu1.rds",
      urls = c(file.path(repo_raw, "inst/extdata/Data_x_simu1.rds")),
      local_fallback = c("example_data/Data_x_simu1.rds")
    ),
    Data_y = load_rds_from_sources(
      "Data_y_simu1.rds",
      urls = c(file.path(repo_raw, "inst/extdata/Data_y_simu1.rds")),
      local_fallback = c("example_data/Data_y_simu1.rds")
    )
  ),
  simu_2 = list(
    Data_x = load_rds_from_sources(
      "Data_x_simu2.rds",
      urls = c(file.path(repo_raw, "inst/extdata/Data_x_simu2.rds")),
      local_fallback = c("example_data/Data_x_simu2.rds")
    ),
    Data_y = load_rds_from_sources(
      "Data_y_simu2.rds",
      urls = c(file.path(repo_raw, "inst/extdata/Data_y_simu2.rds")),
      local_fallback = c("example_data/Data_y_simu2.rds")
    )
  ),
  simu_3 = list(
    Data_x = load_rds_from_sources(

```

```

    "Data_x.rds",
    urls = c(file.path(repo_raw, "example_data/Data_x.rds")),
    local_fallback = c("example_data/Data_x.rds")
  ),
  Data_y = load_rds_from_sources(
    "Data_y.rds",
    urls = c(file.path(repo_raw, "example_data/Data_y.rds")),
    local_fallback = c("example_data/Data_y.rds")
  )
)
)

sim_data <- lapply(sim_data, function(d) {
  d$Data <- list(d$Data_x, d$Data_y)
  d
})

lapply(sim_data, function(d) c(dim(d$Data_x), dim(d$Data_y)))

## $simu_1
## [1] 100 1225 100 1225
##
## $simu_2
## [1] 100 1225 100 1225
##
## $simu_3
## [1] 100 34716 100 34716

```

Simulation Study 1 (simu_1) — tune phi/psi then final result

```

Data_x <- sim_data$simu_1$Data_x
Data_y <- sim_data$simu_1$Data_y
Data <- list(Data_x, Data_y)

simu_1_tuning <- compute_bic_grid(
  Data_x = Data_x,
  Data_y = Data_y,
  q_common = 3,
  q = c(2, 2),
  V = 50,
  gamma = 3,
  rho = 0.95,
  espli1 = 1e-3,
  espli2 = 1e-3,
  phi_seq = seq(1, 3, by = 0.5),
  psi_seq = seq(0.1, 1, by = 0.3)
)

## Running phi = 1 and psi = 0.1
## Running phi = 1 and psi = 0.4
## Running phi = 1 and psi = 0.7
## Running phi = 1 and psi = 1
## Running phi = 1.5 and psi = 0.1

```

```
## Running phi = 1.5 and psi = 0.4
## Running phi = 1.5 and psi = 0.7
## Running phi = 1.5 and psi = 1
## Running phi = 2 and psi = 0.1
## Running phi = 2 and psi = 0.4
## Running phi = 2 and psi = 0.7
## Running phi = 2 and psi = 1
## Running phi = 2.5 and psi = 0.1
## Running phi = 2.5 and psi = 0.4
## Running phi = 2.5 and psi = 0.7
## Running phi = 2.5 and psi = 1
## Running phi = 3 and psi = 0.1
## Running phi = 3 and psi = 0.4
## Running phi = 3 and psi = 0.7
## Running phi = 3 and psi = 1
```

```
simu_1_tuning$BIC_matrix
```

```
##           psi=0.1  psi=0.4  psi=0.7  psi=1
## phi=1      840681.1 840734.6 840788.0 840841.4
## phi=1.5    839232.0 839210.2 839187.8 839210.1
## phi=2      837242.3 837748.1 837687.0 837384.9
## phi=2.5         NA         NA         NA         NA
## phi=3         NA         NA         NA         NA
```

```
simu_1_tuning$optimal_phi
```

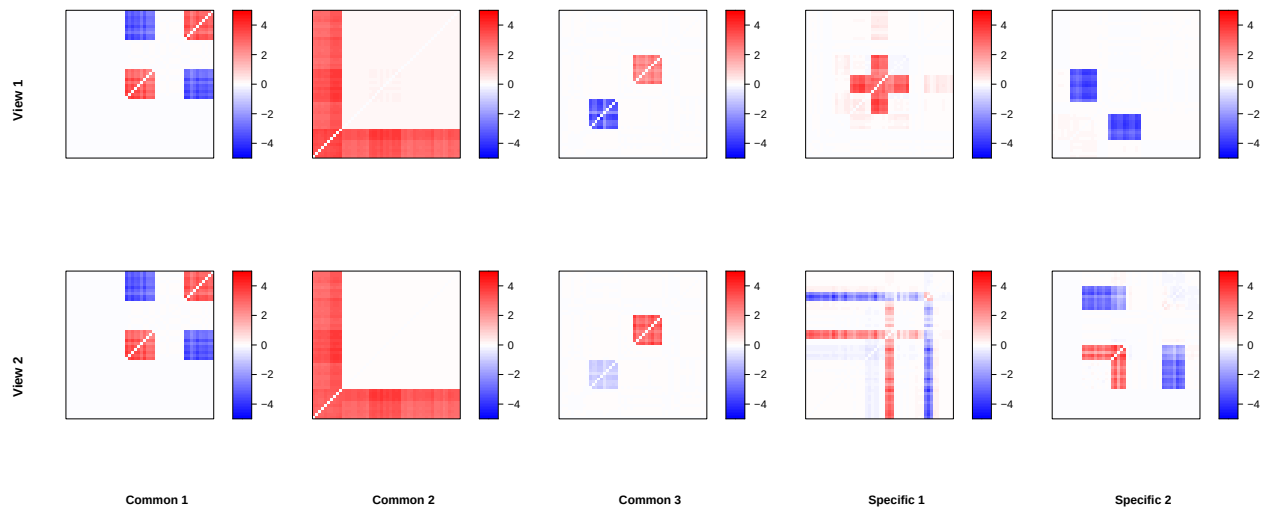
```
## [1] 2
```

```
simu_1_tuning$optimal_psi
```

```
## [1] 0.1
```

```
simu_1_res <- multi_view_decomposition(
  Y = Data,
  q = c(2, 2),
  q_common = 3,
  V = 50,
  penalty = "SCAD",
  phi = simu_1_tuning$optimal_phi,
  psi = simu_1_tuning$optimal_psi,
  gamma = 3,
  rho = 0.95,
  espli1 = 1e-3,
  espli2 = 1e-3,
  MaxIteration = 500,
  silent = T
)
```

```
# simu_1 uses plot_mat-based grid: components 1-3 are common, 4-5 are specific
grid.newpage()
grid.draw(visualize_plot_mat_grid(simu_1_res, V = 50, q_common = 3, q_specific = 2, scale = 4))
```



Simulation Study 2 (simu_2) — tune phi/psi then final result

```
Data_x <- sim_data$simu_2$Data_x
Data_y <- sim_data$simu_2$Data_y
Data <- list(Data_x, Data_y)

simu_2_tuning <- compute_bic_grid(
  Data_x = Data_x,
  Data_y = Data_y,
  q_common = 2,
  q = c(1, 1),
  V = 50,
  gamma = 3,
  rho = 0.98,
  espli1 = 1e-3,
  espli2 = 1e-3,
  phi_seq = seq(1, 3, by = 0.5),
  psi_seq = seq(0.1, 1, by = 0.3)
)
```

```
## Running phi = 1 and psi = 0.1
## Running phi = 1 and psi = 0.4
## Running phi = 1 and psi = 0.7
## Running phi = 1 and psi = 1
## Running phi = 1.5 and psi = 0.1
## Running phi = 1.5 and psi = 0.4
## Running phi = 1.5 and psi = 0.7
## Running phi = 1.5 and psi = 1
## Running phi = 2 and psi = 0.1
## Running phi = 2 and psi = 0.4
## Running phi = 2 and psi = 0.7
## Running phi = 2 and psi = 1
## Running phi = 2.5 and psi = 0.1
## Running phi = 2.5 and psi = 0.4
## Running phi = 2.5 and psi = 0.7
## Running phi = 2.5 and psi = 1
```

```

## Running phi = 3 and psi = 0.1
## Running phi = 3 and psi = 0.4
## Running phi = 3 and psi = 0.7
## Running phi = 3 and psi = 1

simu_2_tuning$BIC_matrix

##           psi=0.1  psi=0.4  psi=0.7  psi=1
## phi=1    587397.3 587403.0 587408.7 587414.5
## phi=1.5  586451.2 586456.9 586462.7 586468.5
## phi=2    585263.3 585269.1 585275.0 585280.8
## phi=2.5  661116.3 661123.6 661127.0 661130.4
## phi=3    624575.5 624579.3 624583.1 624587.0

simu_2_tuning$optimal_phi

## [1] 2

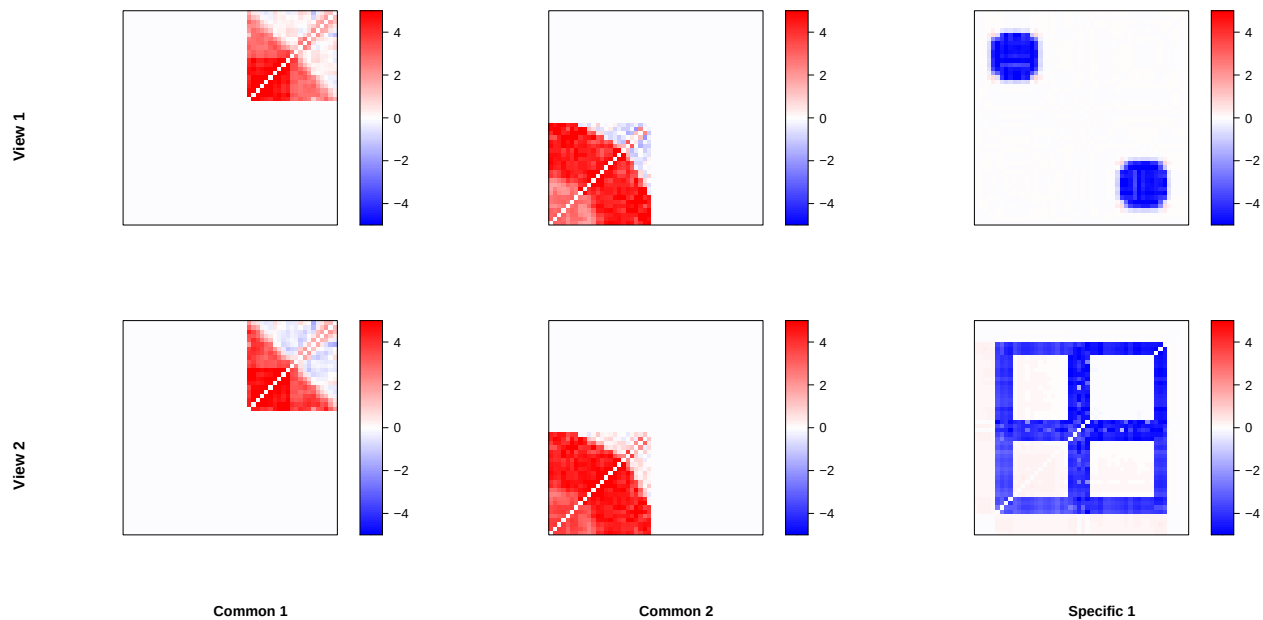
simu_2_tuning$optimal_psi

## [1] 0.1

simu_2_res <- multi_view_decomposition(
  Y = Data,
  q = c(1, 1),
  q_common = 2,
  V = 50,
  penalty = "SCAD",
  phi = simu_2_tuning$optimal_phi,
  psi = simu_2_tuning$optimal_psi,
  gamma = 3,
  rho = 0.98,
  espli1 = 1e-3,
  espli2 = 1e-3,
  MaxIteration = 5000,
  silent = T
)

# simu_2 uses plot_mat-based grid: components 1-2 are common, 3 is specific
grid.newpage()
grid.draw(visualize_plot_mat_grid(simu_2_res, V = 50, q_common = 2, q_specific = 1, scale = 6))

```



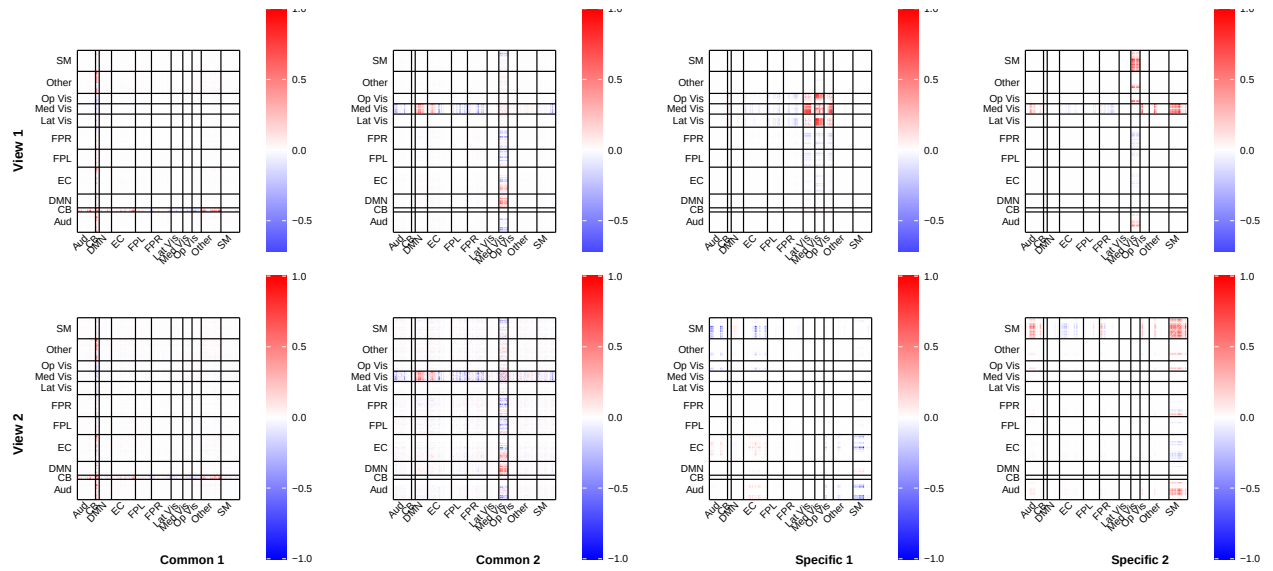
Simulation Study 3 (simu_3) — show results only (no tuning)

```
Data_x <- sim_data$simu_3$Data_x
Data_y <- sim_data$simu_3$Data_y
Data <- list(Data_x, Data_y)
```

```
simu_3_res <- multi_view_decomposition(
  Y = list(Data_x, Data_y),
  q_common = 2,
  q = c(2, 2),
  V = 264,
  penalty = "L1",
  phi = 0.5,
  psi = 0.5,
  gamma = 3,
  rho = 0.95,
  espli1 = 1e-3,
  espli2 = 1e-3,
  MaxIteration = 500,
  silent = T
)
```

```
# simu_3 annotation: components 1-2 are common, 3-4 are specific
```

```
grid.newpage()
grid.draw(visualize_common_specific(simu_3_res, V = 264, q_common = 2, q_specific = 2))
```

```
## [1] "number of annotations: 1"
## [1] "number of annotations: 1"
## [1] "number of annotations: 1"
## [1] "number of annotations: 1"
## [1] "number of annotations: 1"
## [1] "number of annotations: 1"
## [1] "number of annotations: 1"
```